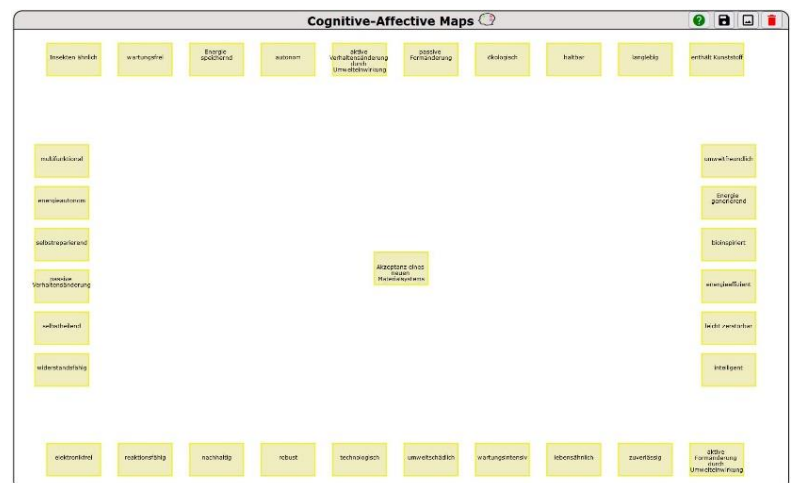


Summary basal attributes article

Main Message. The primary objective of the article is to determine how specific fundamental attributes, when systematically combined, are perceived—either negatively or positively (assessing valence)—in terms of accepting new material systems. This approach is tested through the evaluation of multiple textual descriptions of future material systems that are generated automatically.

Study Design. We are planning a study divided into several successive steps:

1. reduce and translate existing basal attribute list; first reduce it independently (without surveying livMatS researchers); make sure that no too generic basal attributes are included (like "modern", "innovative", ...), which cannot be related to material systems 
2. within a world café the basal attribute list will be evaluated by experts; for each attribute ask (a) about importance for livMatS, (b) about valence [-3,3], further after evaluation participants can freely add further attributes (+ indicate valence) and in additional questions asked about
 - a. relevance of specific basal attribute for Research area (A, B, C, D)
 - b. missing important basal attribute for Research area (A, B, C, D)
 - c. Further livmats publication were automatically scanned for relevant adjectives 
3. main study with laypeople, here the respondents get the task to connect 32 basal attributes to the neutral “Akzeptanz eines neuen Materialsystems” concept in the center of the CAM and change the valence of the single basal attributes; further multiple subsequent questions are surveyed 
4. Using (a) Large Language Model to generate textual descriptions of material systems based on 3. and (b) test textual descriptions by a scenario-based approach



Part I: create a list of basal attributes



(a) Conducted 14 guided expert interviews to create an initial list of basal attributes



(c) Got ratings of 21 experts for (a) the relevance of remaining basal attributes and (b) possible missing's



(b) Reduced initial lists by three raters



(d) Automatically scanned all *livMatS* for possible relevant adjectives using part-of-speech tagging

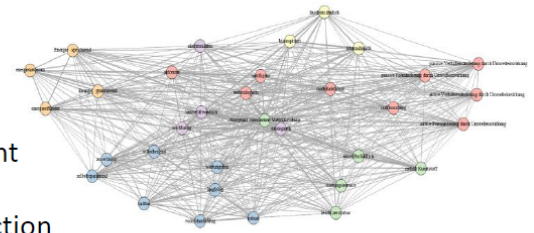
Part II: Cognitive-Affective Maps study



Conducted large-scale CAM study with N=192 laypersons

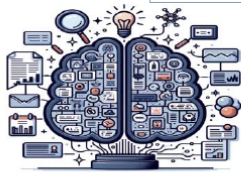


Applied analyses on different levels of the CAMs, using especially community detection algorithms



Identified clusters of basal attributes surrounding the central concept

Part III: use LLMs to create textual descriptions



Applied LLMs to create textual descriptions of future material systems using identified clusters



Scenario #1:
Dave Smith is developing an advertisement for a new housing development his firm is about to start. The development is located in a low area which has flooded in the past. The company has recently done some work to reduce the danger of flooding in the future. In the preliminary advertisement, Smith has included a statement indicating that the firm has solved the flooding problem. The fact is that if a flood occurs, the houses are still likely to have up to a foot of water standing in the yards.

Positive description

Scenario #2:
Dave Smith is developing an advertisement for a new housing development his firm is about to start. The development is located in a low area which has flooded in the past. The company has recently done some work to reduce the danger of flooding in the future. In the preliminary advertisement, Smith has included a statement indicating that the firm has solved the flooding problem. The fact is that if a flood occurs, the houses are still likely to have up to a foot of water standing in the yards.

Negative description



Used ratings of X experts to check the plausibility of textual descriptions



Conducted large-scale study with N=X laypersons to get the acceptance of differently described material systems